Incentives and COVID-19 Vaccinations Replication Project

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Contents

1 Original Study

The paper "Incentives Can Spur COVID-19 Vaccination Uptake" by Heike Klüver, Felix Hartmann, Macartan Humphreys, Ferdinand Geissler, and Johannes Giesecke examines whether policy incentives can increase individuals' willingness to receive a COVID-19 vaccine.

Written during the COVID-19 pandemic, the study addresses the problem of vaccine hesitancy, which threatened efforts to achieve high vaccination coverage and control the spread of the virus.

The authors conducted a large survey experiment with approximately 20,500 participants in Germany. Respondents were presented with different vaccination policy scenarios that varied across policy attributes such as financial incentives, expanded freedoms for vaccinated individuals, and easier access to vaccines through local doctors.

The results suggest that each of these policies can modestly increase willingness to vaccinate, particularly among individuals who are undecided rather than strongly opposed. The authors conclude that combining multiple incentives and improving vaccine access may be an effective strategy for policymakers seeking to increase vaccination rates.

2 Methods

The study analyzes data from a large survey experiment using regression analysis to evaluate how policy incentives influence individuals' willingness to receive a COVID-19 vaccine.

Participants were asked to evaluate hypothetical vaccination scenarios that differed across several policy characteristics, including financial incentives, access to vaccination through local physicians, and expanded freedoms for vaccinated individuals.

The dependent variable measures each respondent's stated likelihood of choosing vaccination under the given scenario. Linear regression models estimate the relationship between vaccination willingness and the randomly assigned policy attributes, allowing the marginal effect of each incentive to be identified.

3 Data

The analysis was conducted in R using R Markdown. The workflow was organized across three scripts:

- prep data: prepares and cleans the dataset used in the analysis
- main results: estimates baseline regression models and produces the main results tables
- het effects: estimates heterogeneous treatment effects for different subgroups

All scripts are compiled through a master document called `index.Rmd`

```
rmarkdown::render("index.Rmd")
```

This procedure executes each script sequentially and reproduces the full analysis pipeline.

4 Original Model

Figures 1, 2, and 3 are based on regression and machine learning models estimating the effects of vaccination incentives.

The analysis first estimates first-difference regressions where the change in vaccination willingness between two scenarios is regressed on changes in assigned treatments: negative incentives, financial incentives of 25 or 50 euros, and improved access through local doctors.

Equivalent individual fixed-effects regressions are also estimated using a stacked dataset with individual indicators to control for unobserved time-invariant characteristics.

```
lm_robust(
  outcome_diff ~ negative_diff + financialA_diff + financialB_diff +
      transaction_diff,
  data = df, se_type = "stata") %>%
  tidy

lm_robust(
  outcome ~ negative + financialA + financialB + transaction,
  data = df_stacked, se_type = "stata", fixed_effects = ~ID) %>%
  tidy

our_analysis <- function(dff)
  lm_robust(
    outcome_diff ~ negative*financialA*transaction + negative*
       financialB*transaction,
    data = dff %>%
      mutate(
        negative = negative_diff - mean(negative_diff),
        financialA = financialA_diff - mean(financialA_diff),
        financialB = financialB_diff - mean(financialB_diff),
        transaction = transaction_diff - mean(transaction_diff)
      ),
    se_type = "stata")

statuses <- unique(df$status)

model_list <- lapply(statuses, function(j) our_analysis(df %>% dplyr
   ::filter(status == j)))

names(model_list) <- statuses

model_list$All <- our_analysis(df)

by_status <-
  lapply(model_list, function(model) tidy(model)) %>%
  bind_rows(.id = "status") %>%
  mutate(status = factor(status, c("All", statuses)))
```

```
policy_experiments <- list(
  experiment_1 =
    df_stacked %>%
    mutate(
      Z = negative * financialB * transaction,
      zip = (1-negative) * (1-financialA) * (1-financialB) * (1-
         transaction)
    ) %>%
    dplyr::filter((Z==1) | (zip==1)) %>%
    group_by(Z) %>%
    mutate(N = n()) %>% ungroup() %>%
    group_by(status, Z) %>%
    summarize(n = n()/mean(N), share_pop_vaccinated = n()*mean(
       outcome)/mean(N)),

  experiment_2 =
    df_stacked %>%
    dplyr::filter(financialA == 0 & negative == 1 & transaction ==
       1) %>%
    group_by(financialB) %>%
    mutate(N = n()) %>% ungroup() %>%
    group_by(status, financialB) %>%
    summarize(n = n()/mean(N), share_pop_vaccinated = n()*mean(
       outcome)/mean(N)) %>%
    arrange(financialB) %>% mutate(Z = financialB)

) %>% bind_rows(.id = "experiment") %>%
  mutate(
    Z = (experiment == "experiment_1") * Z +
      (experiment == "experiment_2") * (Z + .5),
    Z = factor(Z, c(0, .5, 1), c("No␣incentives", "Local␣doctors␣+␣
       Freedoms", "All␣incentives"))
  ) %>%
  dplyr::filter(!is.na(Z))

fig_2_ABC <- ggarrange(
  lm_plot + theme(plot.margin=unit(c(1,1,1.5,1.2)*.6,"cm")),
  ggarrange(
    age_plot + ylim(-.03, .13) + theme(plot.margin=unit(c
       (1,1,1.5,1.2)*.6,"cm")),
    distance_plot + ylab("") + ylim(-.03, .13) + theme(plot.margin=
       unit(c(1,1,1.5,1.2)*.6,"cm")),
    labels = c("B", "C"),
    ncol = 2, common.legend = TRUE, legend="bottom"
  ),
  nrow = 2, labels = "A"
)
```

```
lm_robust(
  outcome_diff ~ negative_diff + financialA_diff + financialB_diff +
      transaction_diff,
  data = df, se_type = "stata"
)
```

4.1 Figure 1: Estimated Effects on Vaccination Probability

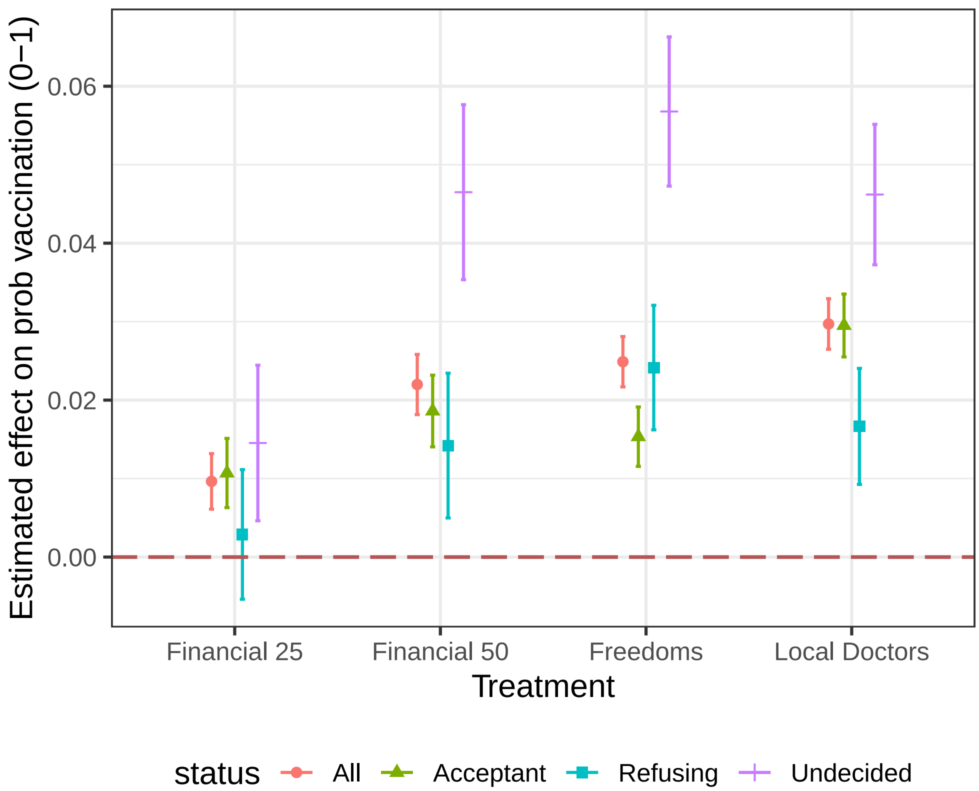

Figure 1: Estimated effects on vaccination probability

Figure 1 shows the estimated treatment effects of different incentives on the probability of vaccination. Financial incentives increase vaccination intentions by roughly three to five percentage points, with the €50 incentive producing a slightly larger effect than the €25 incentive.

Freedom-related incentives increase vaccination probability by roughly two to three percentage points, while access through local doctors produces the largest estimated effect, around five to six percentage points.

The treatment effects are heterogeneous across groups. Individuals who are undecided respond most strongly to incentives, while those already refusing vaccination show little change.

```
figure_1A <- by_status %>%
  filter(term %in% c("negative", "financialA", "financialB", "
     transaction")) %>%
  filter(status != "Vaccinated") %>%
  mutate(Treatment = factor(term, treatment_levels, treatment_labels
     )) %>%
  ggplot(aes(Treatment, estimate, color = status, shape = status)) +
  geom_point(position = position_dodge(width = 0.3)) +
  geom_errorbar(aes(ymin = conf.low, ymax = conf.high),
                position = position_dodge(width = 0.3), width = 0.1)
                    +
  geom_hline(yintercept = 0, linetype = "longdash", size = 0.75,
     colour = "#B55555") +
  theme_bw(base_size = 14) +
  ylab("Estimated effect on prob vaccination (0-1)") +
  theme(legend.position = "bottom")
```

4.2 Figure 2: Predicted Share of Population Vaccinated

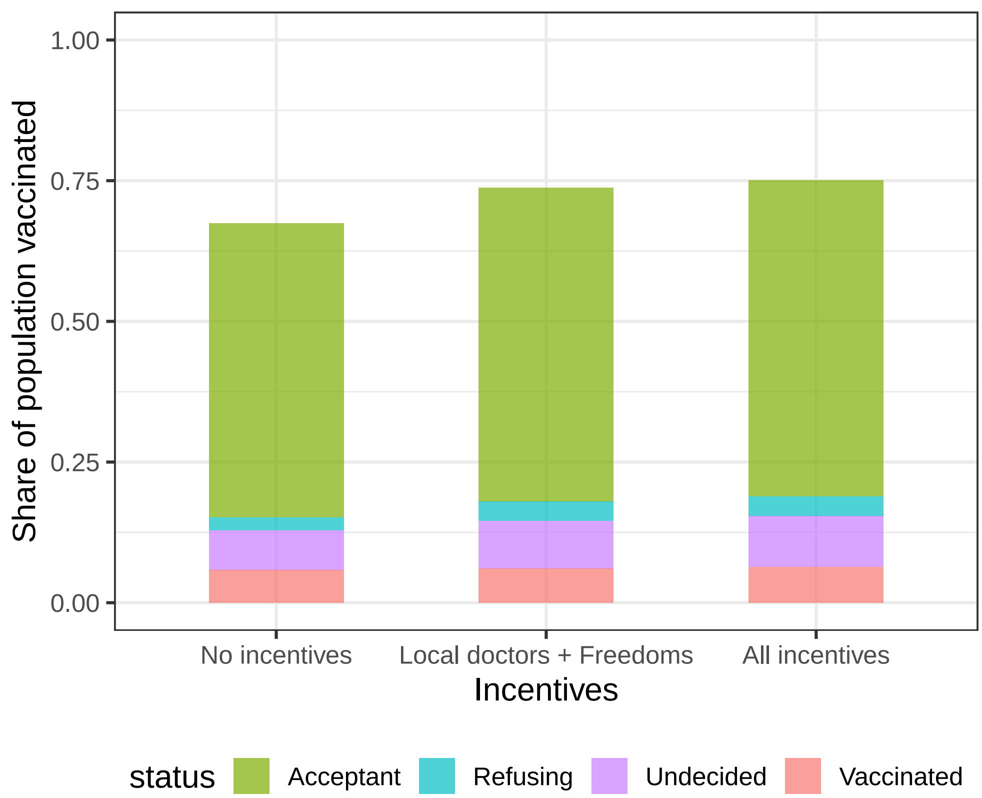

Figure 2: Predicted share of the population vaccinated under different incentive scenarios

Figure 2 presents simulated vaccination outcomes under alternative policy scenarios. In the baseline scenario with no incentives, approximately 50–55% of the population is vaccinated.

Introducing local doctor messaging together with freedom incentives increases the vaccinated share to roughly 60–65%. When all incentives are implemented simultaneously, the predicted vaccination rate rises further to approximately 70–75% of the population.

These increases occur primarily through a reduction in the number of undecided individuals rather than large shifts among those strongly refusing vaccination.

```
figure_1B <- policy_experiments %>%
  ggplot(aes(fill = status, y = share_pop_vaccinated, x = Z)) +
  geom_bar(position = "stack", stat = "identity", width = 0.5, alpha
      = 0.7) +
  scale_fill_manual(values = group.colors) +
  ylab("Share␣of␣population␣vaccinated") +
  xlab("Incentives") +
  theme_bw(base_size = 14) +
  ylim(0:1) +
  theme(legend.position = "bottom")
```

4.3 Figure 3: Heterogeneous Treatment Effects of Vaccination Incentives

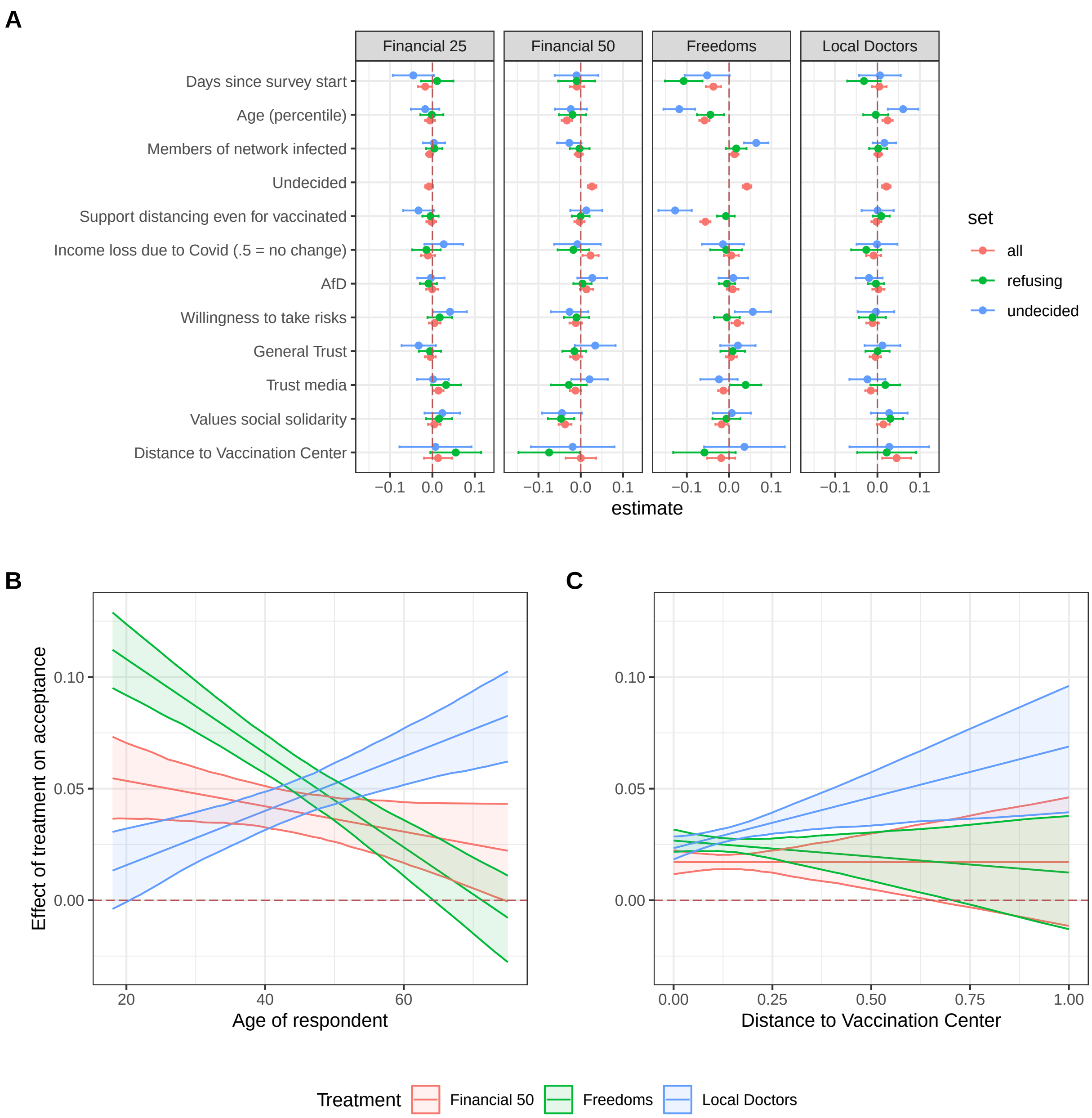

Figure 3: Estimated heterogeneous treatment effects of vaccination incentives across individual characteristics

Figure 3 presents heterogeneous treatment effects of vaccination incentives estimated using a causal forest model. The analysis identifies which covariates are most predictive of variation in treatment effects, including age, prior vaccine hesitancy, socio-economic indicators, and attitudinal measures such as trust in media and risk preferences.

Predicted conditional average treatment effects vary across individuals and demographic characteristics, demonstrating that the impact of policy incentives differs across population groups.

```
fig_2_ABC <- ggarrange(
  lm_plot + theme(plot.margin = unit(c(1,1,1.5,1.2) * 0.6, "cm")),
  ggarrange(
    age_plot + ylim(-0.03, 0.13) +
      theme(plot.margin = unit(c(1,1,1.5,1.2) * 0.6, "cm")),
    distance_plot + ylab("") + ylim(-0.03, 0.13) +
      theme(plot.margin = unit(c(1,1,1.5,1.2) * 0.6, "cm")),
    labels = c("B", "C"),
    ncol = 2,
    common.legend = TRUE,
    legend = "bottom"
  ),
  nrow = 2,
  labels = "A"
)

fig_2_ABC
```

5 Modifications to the Study

To explore heterogeneity in treatment effects, the original regression model was modified to include interaction terms between each treatment variable and normalized age (`age2`). The original model assumed additive effects, meaning that each incentive had the same effect for all respondents.

The modified specification allows treatment effects to vary with age:

$$outcome\_diff \sim treatment + treatment \times age2$$

The interaction term captures how the marginal effect of each treatment changes as age increases.

5.1 Interaction Effects with Age

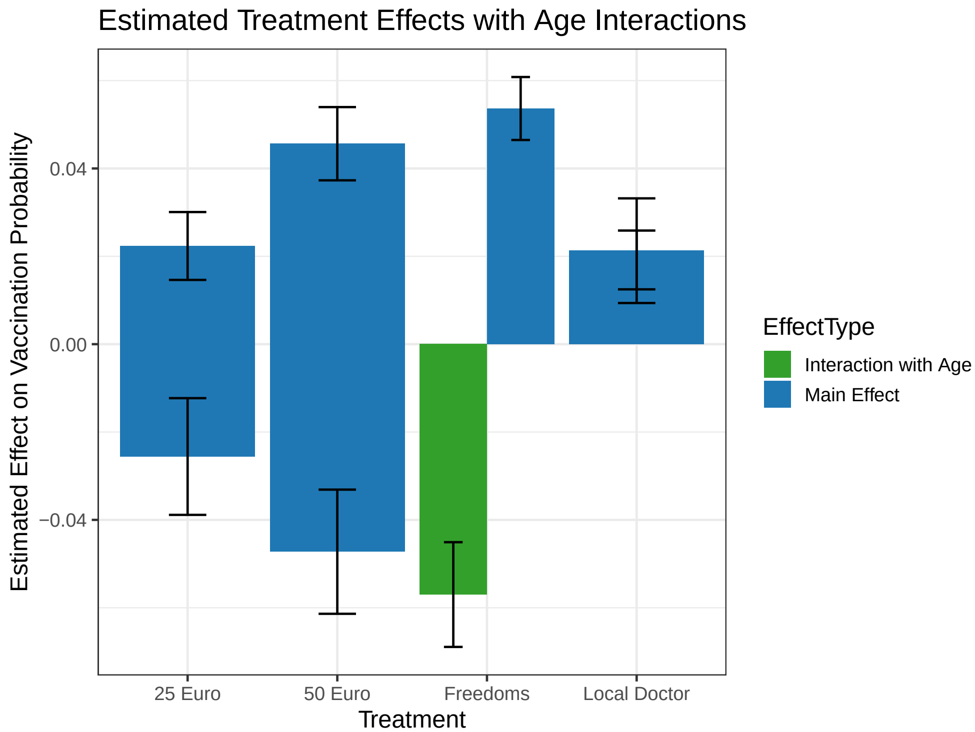

Figure 4: Estimated treatment effects including interactions with age

The results indicate that the effectiveness of incentives differs across age groups. For example, the predicted effect of the freedoms incentive is roughly 0.02 for younger respondents but increases to approximately 0.08 for older respondents.

Financial incentives show similar patterns, while the local doctor incentive exhibits relatively little variation across age groups. These findings suggest that vaccination incentives do not affect all individuals equally and that demographic characteristics such as age play an important role in shaping responses to policy interventions.